

CHANGBAI, China

WMO: 543860

Lat: 41.4249N

Lon: 128.1990E

Elev: 2543

StdP: 13.39

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.1	-14.5	-25.8	1.4	-16.9	-22.2	1.8	-13.1	23.6	6.3	19.7	4.8	4.5	40	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	84.5	66.9	81.7	66.1	78.9	64.9	71.1	79.7	69.5	77.0	68.0	75.0	5.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	115.2	75.2	67.1	109.5	73.6	65.6	104.1	72.0	36.8	79.8	35.3	77.1	34.0	75.1	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.6	16.2	13.5	-23.2	89.1	4.2	2.9	-26.2	91.2	-28.6	92.8	-31.0	94.5	-34.0	96.6		
(5)			-23.5	72.7	4.1	1.8	-26.5	74.1	-28.9	75.1	-31.2	76.2	-34.2	77.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	2.7	10.5	25.3	41.3	53.2	62.3	69.0	67.4	56.1	42.4	23.6	6.8		
	DBStd	24.26	8.01	9.30	9.45	8.31	6.57	5.02	4.32	5.12	6.04	7.58	10.37	8.99		
	HDD50	6064	1467	1106	766	279	44	1	0	0	16	253	792	1340		
	HDD65	9943	1932	1526	1231	712	368	106	14	33	272	702	1242	1805		
	CDD50	1873	0	0	0	17	142	371	589	540	198	16	0	0		
	CDD65	280	0	0	0	0	2	26	139	109	4	0	0	0		
	CDH74	2120	0	0	0	8	128	369	927	645	43	0	0	0		
	CDH80	443	0	0	0	0	22	65	222	133	1	0	0	0		
(14) Wind	WSAvg	5.0	4.4	5.0	6.4	6.5	6.2	4.9	4.3	4.2	4.5	4.9	4.6	4.2		
(15) Precipitation	PrecAvg	25.5	0.3	0.4	0.6	1.3	2.7	3.9	6.5	4.6	2.4	1.1	1.0	0.6		
	PrecMax	33.5	0.8	1.5	1.2	2.7	6.0	6.9	12.9	8.5	5.6	2.2	2.8	1.9		
	PrecMin	16.8	0.0	0.1	0.1	0.4	1.0	1.0	1.9	1.1	0.3	0.1	0.2	0.1		
	PrecStd	4.1	0.2	0.3	0.3	0.6	1.1	1.6	2.5	1.9	1.5	0.6	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.7	40.8	57.4	74.8	82.9	85.4	89.3	87.5	78.2	69.6	53.2	32.2		
		MCWB	25.5	33.1	41.7	53.0	56.7	62.3	69.8	70.5	62.5	52.0	42.2	27.9		
	2%	DB	23.7	35.4	49.8	68.2	77.5	81.7	85.1	83.4	74.6	64.6	47.3	28.3		
		MCWB	20.2	29.3	37.8	49.0	55.2	61.4	68.7	68.9	60.4	49.9	38.5	24.9		
	5%	DB	19.8	31.5	44.7	63.0	73.4	78.2	82.2	80.5	71.6	60.4	42.6	24.8		
		MCWB	16.9	26.6	35.1	46.4	53.4	60.9	68.0	68.0	58.6	47.4	36.1	21.8		
	10%	DB	16.1	27.5	40.2	58.0	69.1	74.9	79.2	77.5	68.5	56.3	38.2	21.2		
		MCWB	13.6	23.4	32.6	43.7	52.1	59.9	66.9	66.4	57.1	45.3	33.4	18.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.3	34.0	42.9	54.7	60.9	66.4	73.4	72.9	66.6	54.2	44.0	29.3		
		MCDB	28.7	38.5	54.7	71.1	76.4	77.3	84.3	82.0	73.1	65.1	50.5	31.2		
	2%	WB	20.4	29.9	38.7	50.2	57.9	64.6	71.8	71.0	63.7	51.2	39.7	25.1		
		MCDB	23.3	34.8	48.1	65.0	71.7	75.1	81.1	79.2	70.8	61.3	45.6	28.0		
	5%	WB	16.9	26.8	35.9	47.3	55.6	63.0	70.1	69.6	61.0	48.9	36.8	21.9		
		MCDB	19.6	31.2	43.6	60.7	68.8	73.3	77.9	77.2	68.1	58.4	42.0	24.7		
	10%	WB	13.6	23.3	33.0	44.6	53.4	61.6	68.5	68.3	59.0	46.5	33.4	18.6		
		MCDB	16.1	27.7	39.5	56.7	66.4	71.3	75.8	75.1	65.6	54.7	37.7	21.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.5	23.8	22.0	23.8	24.9	22.1	18.3	18.6	23.9	24.4	19.9	20.0		
		MCDBR	22.3	24.7	27.4	34.0	34.7	29.3	24.2	22.4	27.5	30.8	25.0	20.2		
		MCWBR	19.2	19.3	16.9	17.4	15.0	11.9	9.6	9.2	13.0	16.7	16.5	17.0		
	5% WB	MCDBR	21.7	22.8	25.0	31.3	28.7	23.1	19.0	18.1	21.0	27.4	22.4	19.7		
		MCWBR	18.7	18.2	16.3	16.9	13.5	10.4	8.5	8.1	10.8	15.5	15.5	16.7		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.384	0.460	0.532	0.567	0.584	0.523	0.474	0.421	0.421	0.369	0.329		
	taud		2.089	1.944	1.816	1.742	1.725	1.754	1.989	2.117	2.183	2.051	2.084	2.096		
	Ebn at Noon		252	251	245	237	233	229	242	251	256	236	231	237		
	Edh at Noon		34	47	60	70	73	71	56	47	41	42	35	32		
(44) All-Sky Solar Radiation	RadAvg		785	1098	1422	1542	1659	1592	1484	1454	1352	1058	720	645		
	RadStd		43	52	81	94	148	153	154	126	128	92	73	32		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.63	N/A	-2.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page